# NetraRakshak: A Clinician-Supervised, AI-Assisted Retinal Screening Platform for Diabetic Retinopathy

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***Abstract*— Diabetic retinopathy (DR) is a leading cause of preventable vision loss, and screening programmes in India are limited by the availability of ophthalmologists rather than by imaging. This paper presents NetraRakshak, a multi-tenant software platform that carries a colour fundus photograph from capture to a clinician-signed screening report. Each image first passes an automatic quality gate of ten computer-vision checks; gradable images are then graded on the five-level International Clinical DR scale by an EfficientNet-B0 network with temperature-scaled confidence, an entropy-based uncertainty rule that returns "inconclusive — human review required", and Grad-CAM explanations. An ophthalmologist accepts, modifies or rejects every AI suggestion before a report can be approved. Trained on the APTOS 2019 dataset (3,662 images), the model achieved, on a held-out test set of 524 images, an accuracy of 81.9%, a quadratic weighted kappa of 0.876 and, for referable DR (moderate non-proliferative DR or worse), a sensitivity of 82.0% and a specificity of 96.6% (ROC-AUC 0.981). Temperature scaling reduced the expected calibration error from 12.2% to 6.1%. Inference including Grad-CAM takes about 0.3 s per eye on a two-core CPU. We describe the system architecture, the safety controls, the measured results and their limitations: the system has not been clinically validated and is intended only as a clinician-supervised screening aid.**

***Index Terms*— diabetic retinopathy, fundus photography, deep learning, EfficientNet, explainable AI, Grad-CAM, model calibration, clinical decision support, medical device software**

## I. Introduction

Diabetic retinopathy is a microvascular complication of diabetes that is often asymptomatic until vision is threatened, which makes regular retinal screening of people with diabetes essential. Grading fundus photographs is time-consuming, and in India the number of trained graders and ophthalmologists is small relative to the population with diabetes. Deep learning systems have shown high accuracy for referable DR on curated datasets [1, 2], and an autonomous system has been cleared for primary-care use in the United States [3]. Translating such models into routine hospital practice, however, requires more than a classifier: image quality control, patient consent, an auditable clinician workflow, data isolation between institutions, reporting and regulatory documentation.

NetraRakshak addresses this gap as a complete software platform. Its contributions are:

- an end-to-end, multi-hospital screening workflow — registration, consent, fundus upload, automatic quality assessment, AI analysis, ophthalmologist review with electronic signature, PDF reporting and analytics;
- a gated model registry that only serves models with verified weights and recorded held-out metrics, keeps the AI output immutable and separate from the clinician's interpretation, and routes uncertain or unsupported cases to human review;
- a DR grading model trained, calibrated and evaluated with a reproducible pipeline, with all reported metrics computed automatically from held-out predictions;
- an analysis of the system's limitations and of the regulatory path for medical device software in India.

## II. Related Work

Gulshan *et al.* [1] trained an Inception-v3 network on about 128,000 graded images and reported an area under the ROC curve of about 0.99 for referable DR on two validation sets. Ting *et al.* [2] validated a system on multi-ethnic populations and showed the importance of external validation across cameras and ethnicities. Ruamviboonsuk *et al.* [4] compared a deep learning system with graders in a national screening programme. Abràmoff *et al.* [3] conducted a pivotal trial of an autonomous AI system in primary care, reporting 87.2% sensitivity and 90.7% specificity for more-than-mild DR. In India, Rajalakshmi *et al.* [5] evaluated AI on smartphone-based fundus photography. Public datasets include EyePACS, APTOS 2019 [6] and the Indian IDRiD [7]. For interpretability, Grad-CAM [14] highlights image regions that drive a prediction, and temperature scaling [13] is a simple, effective method for calibrating neural-network confidence. NetraRakshak builds on these methods but focuses on the surrounding system needed for supervised clinical use.

## III. System Design

### A. Intended Use and Design Principles

NetraRakshak is intended to assist qualified ophthalmologists in screening colour fundus photographs of adults with diabetes for signs of DR. It is not an autonomous diagnostic device, does not diagnose diabetes itself, and does not assess other condi-

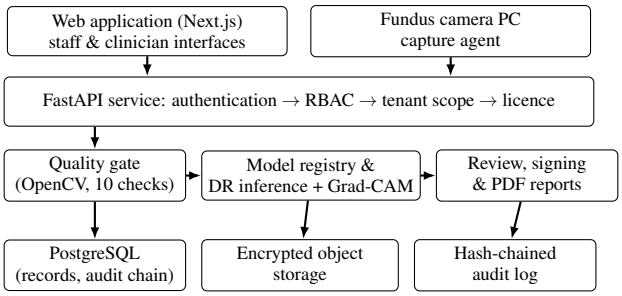



**Figure 1:** NetraRakshak architecture. Every request is authenticated, authorised by role and scoped to one hospital.

tions (glaucoma, age-related macular degeneration and hypertensive retinopathy modules are defined but deliberately inactive). Three principles follow: (i) the AI never has the final word; (ii) no result is produced from an unsuitable image, an unvalidated model or without consent; and (iii) every action is attributable.

### *B. Architecture*

Fig. 1 shows the architecture. The web application is built with Next.js and TypeScript; the API with FastAPI, Pydantic and SQLAlchemy on PostgreSQL; the AI layer with PyTorch, OpenCV and scikit-learn; and reports with ReportLab. Retinal images, explainability maps and reports are encrypted by the application (Fernet: AES-128-CBC with HMAC-SHA256) before being written to private object storage. A reference capture agent uploads images exported by fundus-camera software.

### *C. Workflow*

A receptionist registers the patient and records consent separately for screening, AI analysis, data storage and research. A technician starts a screening session and uploads right- and left-eye photographs (JPEG, PNG or TIFF). The quality gate runs immediately. The AI step then runs on each acceptable image, after which the case enters a review queue ordered by priority: AI-suggested referable cases first, then cases the AI could not resolve. The ophthalmologist inspects the images in a viewer with zoom, pan, side-by-side eyes and Grad-CAM overlays, records a final grade per eye, accepts, modifies or rejects the AI suggestion, recommends referral or recapture, and signs by re-entering their password and attesting to the review. Only then can a report marked "clinician approved" be generated; earlier reports carry a "draft" watermark. Signed sessions are locked.

### *D. Multi-Tenancy and Security*

The platform serves multiple hospitals with five roles (platform administrator, hospital administrator, ophthalmologist, screening technician, receptionist) and a role–permission matrix. Every query on patient data is filtered by the user's hospital, and records of other hospitals return "not found" rather than "forbidden". Platform administrators manage plans and licences but cannot read patient records. Passwords are hashed with bcrypt; accounts lock after repeated failures; access tokens are short-lived and bound to revocable server-side sessions. An append-only audit log records logins, views, uploads, AI runs, reviews, signatures and downloads, each entry chained to the previous one with SHA-256 so that tampering is detectable. The model registry verifies the SHA-256 of model weights against the model card before loading.

**Table 1:** Automatic Image Quality Checks

| Check | Measurement |
|---|---|
| Resolution | short side in pixels ($\geq$512 required) |
| Retinal visibility | fraction of frame covered by the retinal mask |
| Field of view | offset of the mask centroid from the image centre |
| Fundus plausibility | red-channel dominance within the mask |
| Brightness | mean luminance inside the eroded mask |
| Contrast | standard deviation of the green channel |
| Exposure | fraction of clipped and crushed pixels |
| Blur | variance of the Laplacian (CLAHE-enhanced green) |
| Focus | Tenengrad (mean squared Sobel gradient) |
| Artefacts | quadrant illumination spread, specular blobs |

**Table 2:** Images per ICDR Grade in Each Split

| Split | No DR | Mild | Moderate | Severe | PDR | Total |
|---|---|---|---|---|---|---|
| Train | 1281 | 281 | 710 | 134 | 209 | 2615 |
| Validation | 251 | 43 | 156 | 34 | 39 | 523 |
| Test | 273 | 46 | 133 | 25 | 47 | 524 |

### *E. Image Quality Assessment*

Table 1 lists the checks run before any inference, on a resized copy of the image within a segmented retinal mask. Any failure yields "recapture required" with a plain-language reason; three or more borderline results yield "inconclusive". The thresholds are documented defaults that must be calibrated per camera against clinician gradability labels. The checks take about 73 ms per image.

## IV. Deep Learning Method

### *A. Dataset*

The model was trained on the APTOS 2019 Blindness Detection dataset [6]: 3,662 colour fundus images graded 0–4 on the International Clinical DR scale [8]. The data were split into training (2,615), validation (523) and test (524) sets, stratified by grade (Table 2). APTOS does not publish patient identifiers, so grouping was at image level. The data were used under the competition terms accepted by the account holder; commercial use has not been confirmed.

### *B. Preprocessing and Augmentation*

Each image is cropped to the retinal field, padded to a square, resized to $384 \times 384$ pixels and normalised for illumination with Graham's local-average subtraction [9], $I' = 4I - 4\,G_\sigma * I + 128$ with $\sigma$ equal to one-thirtieth of the image size, then standardised with ImageNet statistics. The same code path is used for training and inference. Training augmentation consists of random horizontal and vertical flips, rotation by up to $180^\circ$, scaling by 0.92–1.10 and mild brightness and contrast changes.

**Table 3:** Test-Set Performance (n = 524)

| Metric | Value |
|---|---|
| Accuracy (5-class) | 81.9% |
| Quadratic weighted kappa | 0.876 |
| Macro precision / recall / F1 | 68.0% / 67.8% / 67.0% |
| Weighted F1 | 82.1% |
| Macro ROC-AUC (one-vs-rest) | 0.917 |
| Referable DR sensitivity | 82.0% (168/205) |
| Referable DR specificity | 96.6% (308/319) |
| Referable DR PPV / NPV | 93.9% / 89.3% |
| Referable DR ROC-AUC | 0.981 |
| Expected calibration error, before / after scaling | 12.2% / 6.1% |

**Table 4:** Per-Class Test Performance

| Grade | Sens. | Spec. | Prec. | F1 | AUC |
|---|---|---|---|---|---|
| No apparent DR | 99.3% | 96.8% | 97.1% | 0.982 | 0.998 |
| Mild NPDR | 71.7% | 93.1% | 50.0% | 0.589 | 0.935 |
| Moderate NPDR | 64.7% | 93.1% | 76.1% | 0.699 | 0.946 |
| Severe NPDR | 44.0% | 95.8% | 34.4% | 0.386 | 0.810 |
| Proliferative DR | 59.6% | 98.7% | 82.4% | 0.691 | 0.895 |

### C. Model and Training

The classifier is EfficientNet-B0 [10] initialised with ImageNet weights [11], with its head replaced by dropout ($p = 0.3$) and a five-way linear layer (4.0 M parameters). Class imbalance was addressed with a class-weighted cross-entropy loss (weights inversely proportional to class frequency) with label smoothing of 0.05. Training used AdamW (learning rate $3 \times 10^{-4}$, weight decay $10^{-4}$) with a one-cycle schedule, a batch size of 24 and mixed precision on a single GPU. Model selection used the validation quadratic weighted kappa [12]; training stopped after 14 epochs, and the best checkpoint was from epoch 8 (validation kappa 0.887).

### D. Calibration, Uncertainty and Explainability

A temperature $T$ was fitted on the validation logits by minimising the negative log-likelihood [13], giving $T = 0.771$. For an image with calibrated probabilities $p_k = \mathrm{softmax}(z/T)_k$, the system reports the top class with confidence $c = \max_k p_k$ and the normalised entropy $H = -\sum_k p_k \log p_k / \log 5$. If $c < 0.60$ or $H > 0.75$, the displayed result is "inconclusive — human review required". Grad-CAM [14] is computed on the final convolutional block for the predicted class and stored as an overlay and a heatmap, which are labelled in the interface and the report as model explainability rather than lesions.

## V. Results

### A. Held-Out Test Performance

All metrics in this section were computed automatically by the training pipeline from the 524 test images, which were not used for training, model selection or calibration (Table 3, Table 4, Fig. 2).

### B. System Verification

The backend has 30 automated tests covering authentication and lockout, role permissions, isolation between hospitals, patient management and consent, upload validation, image quality, AI gating (model unavailable, synthetic or tampered models, poor quality, missing consent, uncertainty), clinician signing, report status and the audit chain. All pass. A browser-based end-to-end test reproduced the full workflow — administrator login, registration, upload of both eyes, quality check, AI grading with Grad-CAM, review and signature, PDF generation and dashboard update — in production configuration on PostgreSQL. On a two-core 2.1 GHz CPU, inference including Grad-CAM had a median latency of 304 ms per eye (90th percentile 331 ms). Fig. 3 shows the clinician review workspace.

## VI. Discussion

### A. Clinical Interpretation

The model distinguishes eyes without DR well (sensitivity 99.3%) and has high specificity for referable disease, but at the default operating point it graded 37 of 205 referable eyes (18%) as non-referable, mostly moderate NPDR graded as mild (Fig. 2). Severe NPDR, with only 25 test images, is the weakest class. These results justify the design choice that every case is reviewed and signed by an ophthalmologist; the model must not be used to rule out disease. Tuning the uncertainty thresholds on validation data could route more moderate-versus-mild ambiguity to the "inconclusive" path.

### B. Limitations

First, APTOS lacks patient identifiers, so both eyes of one person may fall into different splits, which can make the results optimistic. Second, the data come from a single source, and performance on other cameras, populations and dilation protocols is unknown; no external validation (for example on IDRiD [7]) has been performed. Third, the quality gate checks image properties, not whether the content is a gradable retina: in testing, the model gave confident outputs on procedurally generated synthetic images, so an out-of-distribution detector is needed. Fourth, the model does not assess diabetic macular oedema. Fifth, clinical validation has not been carried out.

### C. Regulatory and Ethical Considerations

Software intended for screening or diagnosis is a medical device under India's Medical Devices Rules, 2017. CDSCO's draft guidance on medical device software (October 2025) classifies software by the significance of its output and the seriousness of the condition, requires lifecycle quality management, risk management and clinical evaluation, and introduces an algorithm change protocol for AI model updates [15]. The project includes draft intended-use, risk-management (ISO 14971 [16]), software-requirements, cybersecurity and validation-plan documents and is structured for IEC 62304 [17]. Patient data handling follows the purpose-specific consent and erasure model of the Digital Personal Data Protection Act, 2023 [18]. The platform has not been classified, licensed or certified.

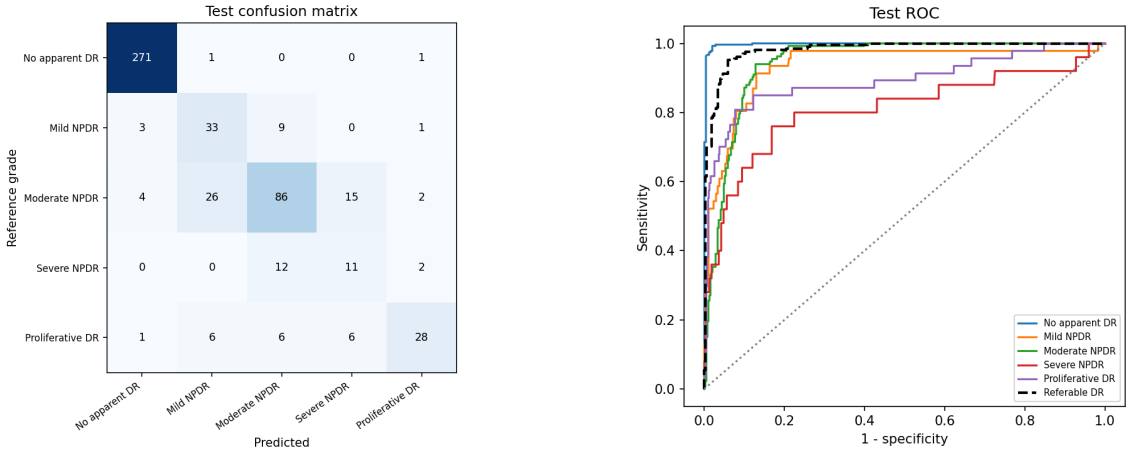



**Figure 2:** Held-out test set (n = 524): confusion matrix (left) and ROC curves per grade and for referable DR (right).

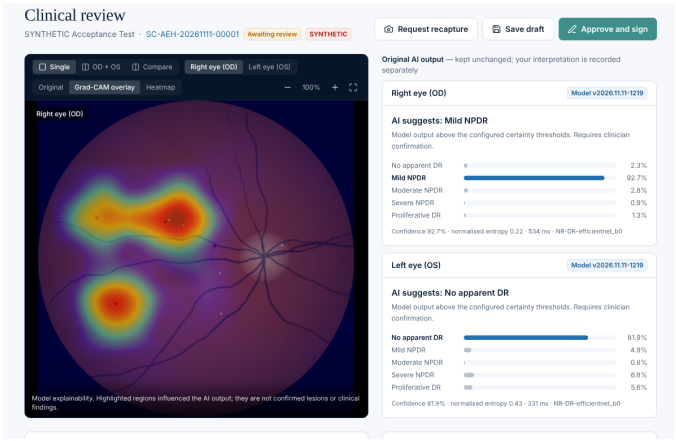



**Figure 3:** Clinician review workspace with a synthetic test patient: Grad-CAM overlay, the unmodified AI output with calibrated probabilities, and per-eye accept, modify or reject decisions.

## VII. Conclusion and Future Work

NetraRakshak shows that an AI grading model can be embedded in a complete, auditable, clinician-supervised screening workflow with quality control, calibrated uncertainty and explainability. The DR model reached a referable-DR sensitivity of 82.0% and specificity of 96.6% on held-out APTOS data. Future work comprises training on licensed, patient-grouped Indian data from several camera types; external validation; an out-of-distribution detector; diabetic macular oedema assessment; tuning the uncertainty operating point; a prospective multi-centre clinical validation study with ethics approval; and a CDSCO submission. Further disease modules will be activated only after their own training and validation.

## Acknowledgment

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